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ScienceDirect

International Journal of Project Management 33 (2015) 711 – 718

International Journal of Project Management

www.elsevier.com/locate/ijproman

Identification and analyses of hidden transaction costs in project dispute resolutions

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Received 24 March 2014; received in revised form 5 August 2014; accepted 28 August 2014
Available online 13 September 2014

Abstract

The hidden transaction costs in project dispute resolutions exert an important influence on the decisions about dispute settlement. In order to obtain the variables and their importance rankings, we firstly developed a thirteen-variable framework of the hidden transaction costs through a literature review and semi-structured interviews. Then we conducted a questionnaire survey to collect the importance of each variable. Based on the results, it is possible to group these variables into five factors – reputation, cooperation and trust, emotion, time, and execution of judgments – by factor analysis. Lack of future cooperation and contractors' reputation damage are the two most important variables, while for owners, project delay is the most severe hidden transaction cost. The findings provide construction practitioners with a deep understanding of the potential hidden loss. Therefore, they will resolve disputes more rationally.
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Keywords: Construction projects; Dispute resolution; Hidden transaction costs

1. Introduction

Conflicts are inevitable because of the divergent interests, demands, or goals of different participants in the construction industry (Cheung and Suen, 2002; Ho and Liu, 2004). If the parties fail to manage a conflict, it may escalate into a dispute (Chong and Zin, 2012; Fenn et al., 1997; Kumaraswamy, 1997). Commonly used methods for dispute resolution consist of negotiation, mediation, arbitration, and litigation. The high cost incurred by arbitration or litigation has become a consensus (Cheung et al., 2010). The above-mentioned costs mean not only the money paid out in the settlement, but also the transaction costs incurred while resolving a dispute (Gebken et al., 2005), which could be considerably high (Li et al., 2012, 2013; Yates, 1999). Reducing such non-value-adding costs is considered as one way to improve the competitiveness of project organizations (Cheung et al., 2000).

Williamson (1985) defined transaction costs as a combination of ex ante and ex post costs. The latter contain "the setup and running costs of the governance structure to which monitoring is assigned and to which disputes are referred and settled." Some scholars have mentioned the transaction costs in dispute resolutions in the literature about transaction costs in construction projects (Hughes et al., 2006; Li et al., 2012, 2013; Mitropoulos and Howell, 2001). For example, Li et al. (2012, 2013) divided transaction costs into pre-contract and post-contract transaction costs, of which dispute resolution costs are one kind of post-contract transaction costs. However, they did not carry out further research into dispute resolution costs. Moreover, Marzouk and Moamen (2009) proposed a framework for estimating the acceptable negotiation amounts. The calculation of the costs incurred by litigation only covered several explicit costs, which can be determined directly, such as court fees and lawyers' fees.

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http://dx.doi.org/10.1016/j.ijproman.2014.08.009
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Yates (1999) pointed out that conflicts and disputes in the construction industry will produce large amounts of costs, including "direct costs (lawyers, consultants, management time and delays to project completions) and indirect costs (degeneration of working relationships, consequences of mistrust between participants and lack of teamwork)." This statement reveals that the transaction costs for dispute resolution comprise direct costs as well as costs that cannot be determined directly. The University of Texas at Austin has conducted several studies of the transaction costs in construction dispute resolutions (Gebken, 2006; Gebken and Gibson, 2006; Gebken et al., 2005), classifying them as direct costs, indirect costs, and hidden costs. We can understand the former two kinds of costs as explicit costs, involving outside and in-house lawyers' fees, outside consultant and expert witness costs, management and staff salary and benefits allocated to supporting the dispute resolution effort, arbitration/mediation/court fees, and so on. They also identified some hidden costs by conducting semi-structured interviews, such as emotional costs, business relationships, and time loss. Notwithstanding, they neither developed the variable framework of the hidden transaction costs for dispute resolution nor analyzed the level of importance of those hidden costs.

In this research, the transaction costs for dispute resolution refer to the non-value-adding costs arising from dispute resolution and incorporate both explicit transaction costs and hidden transaction costs. We consider hidden transaction costs as a kind of hidden loss that can be measured to some extent.

Although it is difficult to measure the hidden loss caused by dispute resolution accurately, hidden transaction costs play an important role in the participants' decisions about receiving or rejecting offers, and may even produce a long-term impact. Regarding the construction industry, which values quantitative results highly, many scholars have provided suggestions for choosing appropriate dispute resolution methods through qualitative analyses (Chan et al., 2006; Cheung, 1999; Cheung and Suen, 2002; Chong and Zin, 2012), and a few scholars have analyzed and quantified the explicit transaction costs of dispute resolution (Gebken, 2006; Gebken and Gibson, 2006; Gebken et al., 2005). Nevertheless, there is a lack of studies that develop a comprehensive variable framework of the hidden transaction costs of dispute resolution to help participants recognize and estimate such hidden transaction costs, probably because of the difficulty of identification. In response to this knowledge gap and to analyze the influence of the hidden transaction costs on the decision-making process, the objectives of this research are: 1) to develop a framework to identify the hidden transaction costs in project dispute resolutions; and 2) to investigate the level of importance of different variables of the hidden transaction costs.

We expect this research to benefit both academia and industrial practitioners in relation to the transaction costs for dispute resolution. It not only lays a foundation for further studies by identifying the variables of the hidden transaction costs, but also provides an insight into the perceived differences of the participants regarding the level of importance of the variables. Construction practitioners can benefit from understanding the hidden transaction costs that different dispute resolution methods incur and thus being able to follow more rational strategies in dispute resolutions. Consequently, unnecessary transaction costs can be minimized.

2. Research methodology

The data collection procedures consisted of three steps, as presented in Fig. 1. The first step was to obtain preliminary variables of the hidden transaction costs in project dispute resolutions through an extensive literature review. Then we performed qualitative research, specifically semi-structured interviews, to improve the framework proposed in step one. We followed a quantitative method, namely a questionnaire survey, to determine the level of importance of the final variables. The following subsections will provide a detailed description of the semi-structured interviews and questionnaire survey as well as the data analyses.

2.1. Semi-structured interviews

A limited amount of literature is retrievable on the hidden transaction costs in the construction industry. Therefore, semi-structured interviews were performed to modify, add, or delete the preliminary variables obtained in the literature review. We contacted five experts (two from contractors, two from owners and one from legal consultants) who specialize in construction project dispute resolution and asked for their agreement to participate in the semi-structured interviews. The experts we selected are practitioners who have more than twenty years' working experience in project management. All of them have participated in construction dispute resolution personally and mastered the detailed information during the process of dispute resolution. As a result, we carried out three interviews with three experts (from contractors, owners and legal consultants, respectively), each of which lasted around one hour. Based on the interview results, we made the following modifications to the proposed framework: 1) two variables were added accordingly: "Expenditure spent on favorable measures taken" and "Difficulty in executing judgments"; 2) the variable "Reputation damage" was divided into two variables from the perspective of contractors and owners, respectively: "Contractors' reputation damage" and "Owners' reputation damage"; 3) every variable was defined and stated in the questionnaire survey for clarity. The new framework of the hidden transaction costs in project dispute resolutions is given in Table 1, with a total of 13 items.

2.2. Questionnaire survey

We conducted a questionnaire survey and designed a questionnaire in the third step. The purpose of the questionnaire was to determine the level of importance of the variables, which facilitated the subsequent analyses of the hidden transaction costs.

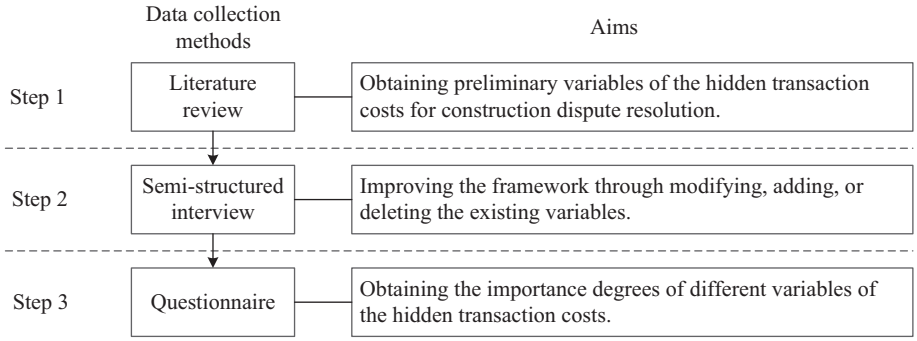


Fig. 1. Data collection procedures.

The questionnaire included two sections. In the first section, we asked the respondents to provide their professional backgrounds, including their years of working experience and the enterprise type. The second section constituted the main body of the questionnaire. It required each respondent to choose his/her perceived importance ratings of the 13 variables on a 7-point Likert scale from 1 (extremely unimportant) to 7 (extremely important), on the basis of their experience. We conducted a preliminary study with 20 samples. Taking into consideration their feedback, we clarified that the respondents should take their enterprises' point of view when considering the loss suffered. Accordingly, we developed the final version of the questionnaire. Between September and November 2013 as well as between May and June 2014, we sent out a total of 400 questionnaires by post or e-mail to practitioners from enterprises derived from a list of the Top 100 contractors, as well as owners, in Mainland China. We

Table 1
The framework of the hidden transaction costs.

Variables	Descriptions	Sources
No. 1 Time loss of claim personnel	In order to settle the dispute, claim personnel cannot carry out other work, which causes time loss.	Gebken, 2006; Hughes et al., 2006; Marzouk and Moamen, 2009; Yates, 1999
No. 2 Delayed recovery of money	The delayed recovery of money incurred by the dispute produces loss (e.g. opportunity loss due to a lack of other investments or an increase in funding costs).	Gebken, 2006
No. 3 Project delay	The project delay caused by dispute resolution brings forth loss.	Gebken, 2006; Gebken and Gibson, 2006; Gebken et al., 2005; Marzouk and Moamen, 2009; Sambasivan and Soon, 2007; Yates, 1999
No. 4 Quality loss of follow-up work	Quality defects of follow-up work caused by dispute resolution generate loss.	Gebken, 2006; Gebken and Gibson, 2006; Gebken et al., 2005
No. 5 and No. 6 Reputation damage	No. 5 The contractors' reputation damage incurred by the dispute results in a decrease in bidding competitiveness (from the contractors' perspective). No. 6 The owners' reputation damage incurred by the dispute results in failure to employ qualified contractors in the future or an increase in completion costs (from the owners' perspective).	Gebken, 2006; Gebken and Gibson, 2006; Gebken et al., 2005; Marzouk and Moamen, 2009; Nieto-Morote and Ruz-Vila, 2012; Cialdini et al., 2004; Yiu et al., 2011
No. 7 Lack of future cooperation	The lack of future cooperation caused by the dispute engenders loss.	Li et al., 2013; Marzouk and Moamen, 2009; Tsai and Chi, 2009
No. 8 Effect on other cooperation	Other ongoing cooperative projects between the two parties can be influenced by the dispute, which brings about loss.	Marzouk and Moamen, 2009
No. 9 Trust damage	Both parties take stricter actions to protect their own interests because of the trust damage caused by the dispute, which gives rise to loss.	Yates, 1999; Zaghloul and Hartman, 2003
No. 10 Reduction in working efficiency of the project	The reduction in working efficiency of the project arising from the dispute results in loss.	Gebken, 2006; Gebken et al., 2005
No. 11 Emotional costs	The bad emotions of personnel caused by dealing with the dispute affect their normal family life and further lead to loss as a result of mistakes at work.	Gebken, 2006
No. 12 Expenditure spent on favorable measures taken	With the aim of resolving the dispute and maintaining an amicable cooperative relationship, the parties may take some favorable measures (e.g. expenses of dining, entertainment, and gifts), which cause loss.	Semi-structured interviews
No. 13 Difficulty in executing judgments	The difficulty in executing judgments gives rise to loss.	Semi-structured interviews

received 233 valid questionnaires, representing an effective response rate of 58.25%.

3. Data analyses and results

3.1. Sample characteristics

We counted the characteristics of the respondents participating in the questionnaire survey. Table 2 presents detailed information on the respondents.

3.2. Importance rankings of the variables

With the data acquired from the questionnaire, we performed a ranking analysis to identify the important variables. We calculated the mean and standard deviation of each variable to represent its level of importance. If two or more variables happened to have the same mean, we considered the one with the lower standard deviation to be more important (Cheung, 1999; Wang and Yuan, 2011). Table 3 shows the ranking results of the 13 variables.

With the purpose of gaining a better understanding of the influence of the different enterprise types on the level of importance of each variable, we divided the data into a contractor group (N = 145) and an owner group (N = 88) for further analyses. Firstly, we conducted Kendall's concordance test to measure the agreement of the respondents' scores within each group. The results showed that the values of chi-square within the contractor group and the owner group were all above the threshold requirement of 26.217, and the levels of significance were all less than 0.01. As a consequence, there was significant agreement amongst the respondents within a group. Secondly, we adopted Spearman's rank correlation test to measure the strength of the correlation between the importance rankings of the variables perceived by the two groups. Construction management research related to ranking analyses has widely used this method (Chan et al., 2011; El-Sayegh, 2008; Odeh and Battaineh, 2002; Wong et al., 2000). The Spearman's rank correlation coefficient was 0.907 and the level of significance was less than 0.01, implying that the importance perception of the variables between the two groups correlated with each other significantly. Moreover, we chose the Mann–Whitney *U* test to identify the particular variables for which the two groups held different views regarding the level of importance. The results indicated that the contractor group assigned higher ratings than the owner group to the importance of "Contractors' reputation damage" (M = 5.035 and 4.636) and "Lack of future cooperation" (M = 5.083 and 4.761), whereas the owner group provided a higher rating on the importance of "Owners' reputation damage" (M = 4.046 and 3.593). Table 4 shows the corresponding statistical results.

We calculated the mean scores of the 13 variables and ranked them in descending order for the contractor group and the owner group, respectively. "Lack of future cooperation" (M = 5.083, SD = 1.493) was the most important hidden transaction cost for the contractors, whereas the owners ranked "Project delay" (M = 4.898, SD = 1.278) in the first place.

Table 2
Sample characteristics.

Characteristic	Number	Percent
Years of working experience		
<5 years	72	30.9
5–9 years	60	25.7
10–19 years	57	24.5
≥20 years	44	18.9
Average	10	
Enterprise type		
Contractor	145	62.2
Owner	88	37.8
State-owned enterprise	212	91.0
Private enterprise	21	9.0

Table 3
Importance rankings of the variables.

Variables	Mean	Standard deviation	Ranking
No. 7 Lack of future cooperation	4.961	1.495	1
No. 5 Contractors' reputation damage	4.884	1.619	2
No. 2 Delayed recovery of money	4.854	1.308	3
No. 3 Project delay	4.854	1.318	4
No. 9 Trust damage	4.511	1.326	5
No. 8 Effect on other cooperation	4.502	1.497	6
No. 13 Difficulty in executing judgments	4.485	1.330	7
No. 10 Reduction in working efficiency of the project	4.189	1.306	8
No. 4 Quality loss of follow-up work	4.094	1.687	9
No. 1 Time loss of claim personnel	4.060	1.410	10
No. 6 Owners' reputation damage	3.764	1.722	11
No. 11 Emotional costs	3.206	1.568	12
No.12 Expenditure spent on favorable measures taken	2.927	1.339	13

3.3. Factor analyses

We conducted exploratory factor analyses to develop the classifications of the variables with the contractor data and the owner data used in the importance rankings, respectively. The results showed that the general classifications were the same for the two groups. For the sake of parsimony, we combined the data together and only reported the results of the factor analysis with the whole data. Firstly, we performed the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and the Bartlett test of sphericity to evaluate the suitability of the data. The KMO value was 0.792, higher than the threshold value of 0.5 (Hair et al., 2006). The Bartlett test of sphericity result was 725.256 and the level of significance was 0.000. Thus, the data were suitable for factor analysis. We chose the commonly used Varimax rotation, which extracted five factors. The fifth factor was also extracted because its eigenvalue was 0.944, which is close to 1. Table 5 presents the final factor loading matrix. In accordance with the meanings of the variables in each classification, we named the factors as follows: reputation (Factor 1), cooperation and trust (Factor 2), emotion (Factor 3), time (Factor 4), and execution of judgments (Factor 5).

Table 4
Statistical results of Kendall's concordance test, Spearman's rank correlation test and Mann–Whitney *U* test.

Statistical tests		All respondent groups	
		Contractor group	Client group
1. Kendall's concordance test			
Actual calculated value of chi-square		382.798	173.465
Critical value of chi-square from table		26.217	26.217
Degree of freedom		12	12
Level of significance		0.000	0.000
2. Spearman's rank correlation test			
Spearman's rank correlation coefficient		0.907	
Level of significance		0.000	
3. Mann–Whitney U test			
Level of significance			
	Contractors' reputation damage	0.025	
	Lack of future cooperation	0.085	
	Owners' reputation damage	0.034	

3.3.1. Factor 1: reputation

Factor 1 comprises "Quality loss of follow-up work," "Contractors' reputation damage," and "Owners' reputation damage."

If the hostility between the parties deteriorates, contractors may control the quality carelessly, or even lower the quality to cut the loss. Quality defects in follow-up work are therefore likely to appear and cause loss. Furthermore, reputation damage gives rise to hidden loss. This kind of loss is mainly reflected in a decrease in bidding competitiveness for contractors. Similarly, owners' reputation will be partly damaged in the case of a high frequency of dispute occurrence. Contractors will take owners' reputation risk into consideration and choose to increase their risk premium based on the general bidding price. Owners' completion costs will rise accordingly. More seriously, contractors will perhaps give up the cooperation opportunity with such owners, making it hard for the owners to employ qualified contractors. However, the hidden loss that reputation damage incurs for owners is much lower than that for contractors, as showed in Table 3.

3.3.2. Factor 2: cooperation and trust

Factor 2 includes "Trust damage," "Lack of future cooperation," and "Effect on other cooperation."

Disputes not only affect the future cooperation between the two parties (see the detailed analyses in the Discussion), but also bring about trust damage. Eventually, they will take stricter measures to supervise and restrict the other party with the aim of protecting their own interests. For example, the owner may designate more people as supervisors to ensure that the contractor complies with the contract rigorously. The supervisory cost will increase as a consequence. In turn, the contractor will probably resort to strategies like claims or variations to squeeze the profits (Cheung and Pang, 2013; Khalfan et al., 2007), in the belief that the pay will not be a fair return. Under this circumstance, the owner should respond to more opportunistic behaviors, resulting in a cost increase. What is more, when disputes occur, both parties try to find out the other party's faults so that they can contradict the demands or requests of their opponent (Gebken, 2006). All such actions taken will produce unnecessary loss. Besides, the parties may cooperate on other projects at the same

Table 5
Factor analysis of the variables.

Variable	Factors				
	1	2	3	4	5
No. 4 Quality loss of follow-up work	**0.804**	−0.071	0.139	0.272	0.018
No. 5 Contractors' reputation damage	**0.699**	0.374	−0.052	0.050	0.094
No. 6 Owners' reputation damage	**0.669**	0.332	0.139	0.039	−0.043
No. 9 Trust damage	−0.093	**0.766**	0.208	0.175	0.197
No. 7 Lack of future cooperation	0.263	**0.752**	0.041	0.069	0.003
No. 8 Effect on other cooperation	0.367	**0.702**	0.182	0.107	−0.050
No. 12 Expenditure spent on favorable measures taken	−0.143	0.018	**0.822**	0.097	−0.027
No. 11 Emotional costs	0.369	0.140	**0.744**	0.006	0.093
No. 10 Reduction in working efficiency of the project	0.159	0.305	**0.594**	0.081	0.200
No. 1 Time loss of claim personnel	0.026	0.045	0.221	**0.767**	−0.163
No. 2 Delayed recovery of money	0.106	0.293	−0.040	**0.686**	0.007
No. 3 Project delay	0.417	−0.019	−0.004	**0.607**	0.292
No. 13 Difficulty in executing judgments	0.032	0.091	0.138	−0.039	**0.927**

time. In this case, the strained relationship due to this dispute is likely to trigger more disputes in the other cooperative projects. The increasingly negative feelings will make both parties lose the patience and willingness to solve problems in a friendly manner, and thus will have a bearing on the economic benefits of the other projects.

3.3.3. Factor 3: emotion

There are three variables in Factor 3, namely "Expenditure spent on favorable measures taken," "Emotional costs," and "Reduction in working efficiency of the project."

"Expenditure spent on favorable measures taken" is a sort of relationship cost. Managerial personnel may take beneficial actions (e.g. dining, entertainment, and gift giving) to reduce hostility and promote amicable dispute resolution, especially for contractors. Some scholars have pointed out that disputes also exert an effect on the personnel's emotions, and further influence their work. Mistakes are likely to occur, which give rise to loss. Nevertheless, our results indicate that the respondents generally believe that the loss caused by these two variables is relatively small, as shown in Table 3. In addition, some personnel (e.g. project managers and auditors) have to perform other duties as well as dispute resolution. It is particularly hard to finish what they should do well due to the lack of sufficient time once they fall into disputes (Gebken, 2006). The working efficiency will be reduced as a result of overburdening.

3.3.4. Factor 4: time

Factor 4 consists of "Time loss of claim personnel," "Delayed recovery of money," and "Project delay."

Both parties have to spend a great deal of time on settling disputes (Hwang, 1998). Time loss is inevitable since claim personnel cannot carry out other work. Sometimes the execution of judgments is put off, due to the imperfection of the district laws and regulations or the weakness of the executive force. It also prolongs the time for dispute resolution. Money has a time value (Nie et al., 2009; Rubinstein, 1982). If the recovery of money is delayed due to disputes, the party who has the right to own the money will lose the opportunity to make other investments or suffer from rising funding costs. Ultimately, the added value of money will disappear. Scheduling is one of the major goals in construction projects (Kog and Loh, 2012). Unresolved disputes imply a hold-up of the payment. Without payment, contractors tend to abstain from working (Ng et al., 2007) or slow down their progress, for the purpose of pressurizing the owners. This will definitely delay projects, as well as producing poor schedule performance. For owners, delay means loss of revenue (Assaf and Al-Hejji, 2006). Hence, owners usually hope that contractors can guarantee good construction progress when disputes occur (Lu and Liu, 2014), and would like to share the profits earned with the contractors. On the condition that contractors are not entitled to an extension of time and suffer project delay, they should pay high delay damages to owners.

3.3.5. Factor 5: execution of judgments

Factor 5 merely contains "Difficulty in executing judgments."

During the dispute resolution process in the Chinese construction industry, mediators often take this variable as an excuse to persuade the two parties to reach an agreement. Many reasons account for the difficulty in executing judgments. For example, local protectionism may happen for the protection of local enterprises; courts can do nothing about judgment debtors' behaviors, such as hiding, transferring, or selling off their subject matter of execution on account of the lack of a credit checking and asset tracking system in China (Blazey and Gillies, 2008; Zhang, 2002). All these will prevent the prevailing party from obtaining the deserved property in whole or in part. Unfamiliarity with the legal system in the project location is one of the causes of disputes in international projects (Chan and Suen, 2005). In this regard, practitioners who participate in international projects located in China are supposed to pay great attention to this hidden transaction cost.

4. Discussion

The importance rankings show that the respondents ranked "Lack of future cooperation" as the most important hidden transaction cost. As a collectivistic country, China attaches great importance to harmony and the maintenance of interpersonal relationships (Friedman et al., 2005; Hwang, 1998; Tsai and Chi, 2009). Dispute escalation may bring about damage to the relationship between the two parties (Rahim, 2001). Besides, it has detrimental effects on the attitude towards future cooperation and reduces the likelihood of working together again. As a result, a hidden loss occurs. For the weaker party in the construction industry, this kind of loss incurred by the lack of future cooperation is much more severe. Owners usually hold greater power in the construction industry (Harmon, 2003; Lu and Hao, 2013). Therefore, contractors, who hold a relatively weaker position, generally pay a higher degree of attention to the loss caused by losing future cooperation with the other party, as showed in the Mann–Whitney U test.

The respondents perceived "Contractors' reputation damage" as the second most important variable. In prequalification documents, owners generally require contractors to fill in their reputation status, covering their claims and litigation history in past projects (Nieto-Morote and Ruz-Vila, 2012). If a contractor frequently fell into litigation or arbitration, and always acted as a losing party, its reputation level would be affected to some extent, decreasing its bidding competitiveness. Under the same conditions, owners will choose other contractors as partners preferentially. In turn, this contractor has to increase its bidding competitiveness by cutting down the bidding price or improving the exclusiveness of technology to win a project. Finally, the bidding cost will increase greatly.

One interesting finding that emerged was that the hidden loss caused by reputation damage varied widely for owners and contractors. This result may be explained by the fact that the Chinese construction industry operates in a buyer's market, implying that the competition among contractors is fierce (Lo et

al., 2007; Lu and Hao, 2013; Zhao et al., 2009). If a contractor's reputation is damaged because of a dispute, it has to spend a large amount of money on enhancing its competitiveness, in order to win the bidding in such a competitive environment. On the contrary, owners hold an advantaged position in the market. The probability and the degree of the hidden loss resulting from reputation damage are relatively low. Moreover, in terms of reputation damage, both parties are likely to emphasize their own loss more. That may account for the outcome that the importance score of "Contractors' reputation damage" was significantly higher for contractors than that for owners, while owners put a higher value on "Owners' reputation damage" than contractors.

5. Conclusion

This research reports on the framework of hidden transaction costs in project dispute resolutions and the level of importance of different variables. To achieve these, we performed a literature review and semi-structured interviews first to identify the variables and develop the framework. Then we conducted a questionnaire survey to determine the level of importance of the hidden transaction costs.

Some conclusions can be drawn from this study: 1) it is possible to divide the thirteen variables identified into five factors, namely reputation, cooperation and trust, emotion, time, and execution of judgments; 2) among all the variables, lack of future cooperation and contractors' reputation damage are the most important; and 3) the hidden loss incurred by owners' reputation damage is relatively small, while that caused by project delay is the most severe for owners.

5.1. Theoretical and practical implications

This study contributes to the literature in three main ways. First, while there is a large body of knowledge regarding the appropriateness of dispute resolution methods in the extant literature, few studies have considered the influence of the hidden transaction costs in project dispute resolutions. By identifying a comprehensive variable framework, we uncover the important role of the hidden transaction costs, which should also be taken into account when resolving a dispute. Second, the findings reveal the specific effect of different variables of the hidden transaction costs for both parties. It is critical for developing a nuanced understanding of how these variables exert different impact on the participants' decision-making process. Third, this research provides more finer-grained analyses of the hidden transaction costs associated with dispute resolution efforts, which extends the body of knowledge on the transaction costs in project dispute resolutions.

The practical implications manifest in two primary aspects. First of all, construction practitioners are expected to make efforts to minimize the useless hidden transaction costs. For example, our study indicates that owners highlight the duration of projects most. They wish to put the projects into operation to obtain a payoff as soon as possible. Hence, expediting the progress to avoid a project delay before resolving disputes appears to be a reasonable approach for contractors. This strategy is conducive to winning the recognition of owners as well as reducing contractors' own hidden loss. Similarly, contractors are suggested to guarantee a good quality performance. A give and take is encouraged, implying that owners had better make some acceptable concessions to recognize the efforts made by contractors. Second, practitioners should estimate possible hidden transaction costs during the dispute resolution process in advance, and then make rational decisions about whether it is worthy to escalate disputes. Generally, negotiation is the most cost-effective choice to resolve disputes. Dispute escalation may result in serious hidden loss, such as lack of future cooperation, reputation and trust damage. In addition, even the prevailing party in the arbitration or litigation may suffer a loss incurred by delayed recovery of money arising from the time-consuming dispute resolution process or the low level of execution efficiency, and may not even be able to obtain the deserved property in whole or in part due to the difficulty in executing judgments. They also have to spend a large amount of money on the explicit expenditure, consisting of lawyers' fees and arbitration/mediation/court fees. This means that the transaction cost for dispute resolution could be tremendous, even exceeding the payoff brought by the judgment result, which leads to a lose-lose situation. It is thus recommended that both parties should balance the potential gain and loss when resolving disputes. Besides, the results show that hidden loss caused by contractors' reputation damage is much more severe than that incurred by owners' reputation damage. From this perspective, contractors should be more careful about inappropriate dispute settlement.

5.2. Limitations and future research

Even though this study makes the above-mentioned contributions, several limitations and future research opportunities should be highlighted. First, the applicability of the results may be limited for the reason that we only collected the empirical data in China. Although the findings have certain implications for dispute resolution in other countries, future studies are expected to undertake the validation. Besides, we merely identified the hidden transaction costs for project dispute resolution and analyzed their impact. Further research could put forward some measurement models to estimate the hidden transaction costs to some extent, in order to provide quantitative support for dispute resolution.

Conflict of interest

We declare that we have no conflict of interest with other people or organizations.

Acknowledgments

The authors would like to acknowledge the support from the National Natural Science Foundation of China (Project Number 71172147 and 71231006), and all the respondents engaged in the questionnaire survey. Additionally, the valuable comments and suggestions from the editor and all the reviewers are highly appreciated.

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